RAJALAKSHMI ENGINEERING COLLEGE

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V2

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|---|---|
| **Started on** | Wednesday, 7 October 2026, 9:11 AM |
| **State** | Finished |
| **Completed on** | Wednesday, 7 October 2026, 9:12 AM |
| **Time taken** | 52 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

**Example 1:**

```
Input: nums = [3,2,3]
Output: 3
```

**Example 2:**

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

**Constraints:**

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

**For example:**

| Input | Result |
|---|---|
| 3<br>3 2 3 | 3 |
| 7<br>2 2 1 1 1 2 2 | 2 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int findMajorityElement(int *nums, int n) {
    int candidate = nums[0];
    int count = 1;

    for (int i = 1; i < n; i++) {
        if (count == 0) {
            candidate = nums[i];
            count = 1;
        } else if (nums[i] == candidate) {
            count++;
        } else {
            count--;
        }
    }

    return candidate;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1 || n <= 0) {
        return 0;
    }
```

```
27
28      int *nums = (int*)malloc(n * sizeof(int));
29      for (int i = 0; i < n; i++) {
30          if (scanf("%d", &nums[i]) != 1) {
31              free(nums);
32              return 0;
33          }
34      }
35
36      printf("%d\n", findMajorityElement(nums, n));
37
38      free(nums);
39      return 0;
40  }
41
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>3 2 3 | 3 | 3 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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